

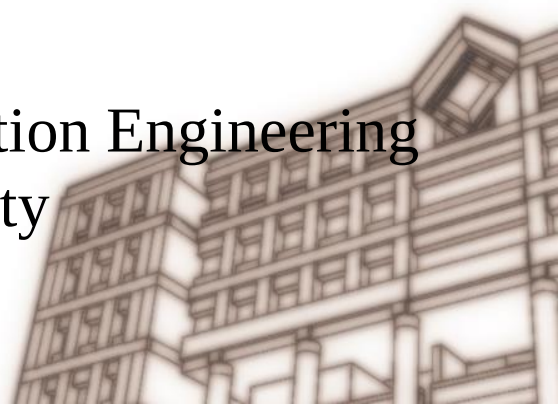
A Modified Optical Flow Estimation Approach

410410004 Pin-Hsiu Lin 林品秀

410410057 Yu-Ren Wang 王昱人

410410086 Chen-Yu Ko 柯宸瑤

Department of Computer Science and Information Engineering
National Chung Cheng University



Outline

- Introduction
- FlowFormer++
- Proposed Approach
- Experimental Results
- Concluding Remarks

Introduction

- Optical flow estimation aims to find the 2D dense motion fields between two video frames.
- Most optical flow estimation methods often rely on local clues, resulting in fragmented motion estimation results.
- In this study, segment anything model (SAM) is employed to cope with the above-mentioned problem. Additionally, the Gaussian constrained layer (GCL) is employed to emphasize local properties and motion affinity.

FlowFormer++

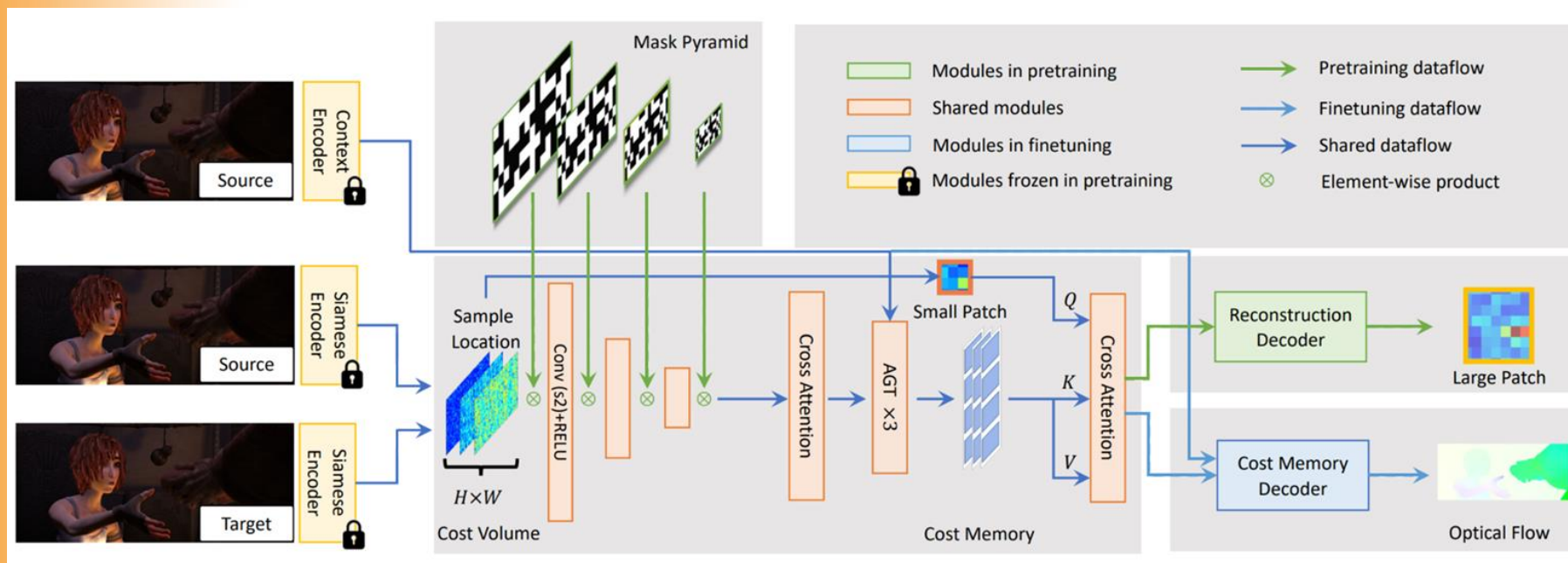


Fig. 1. FlowFormer++ architecture.

FlowFormer++ (cont.)

- FlowFormer++ performs optical flow estimation in three steps:
 1. A 4D cost volume is built for extracted frame features.
 2. A cost volume encoder encodes the 4D cost volume into the cost memory.
 3. A recurrent transformer decoder decodes the cost memory with the source frame context features into optical flows.

Proposed Approach

- The framework of the proposed approach is shown in Fig. 2.
- Each input video frame (source frame) is processed simultaneously by the context encoder of FlowFormer++ and the SAM encoder.
- The outputs of these two encoders are then passed through the pretrained SAMflow architecture containing context fusion module (CFM) and context adaption module (CAM) to extract frame context features.

Proposed Approach (cont.)

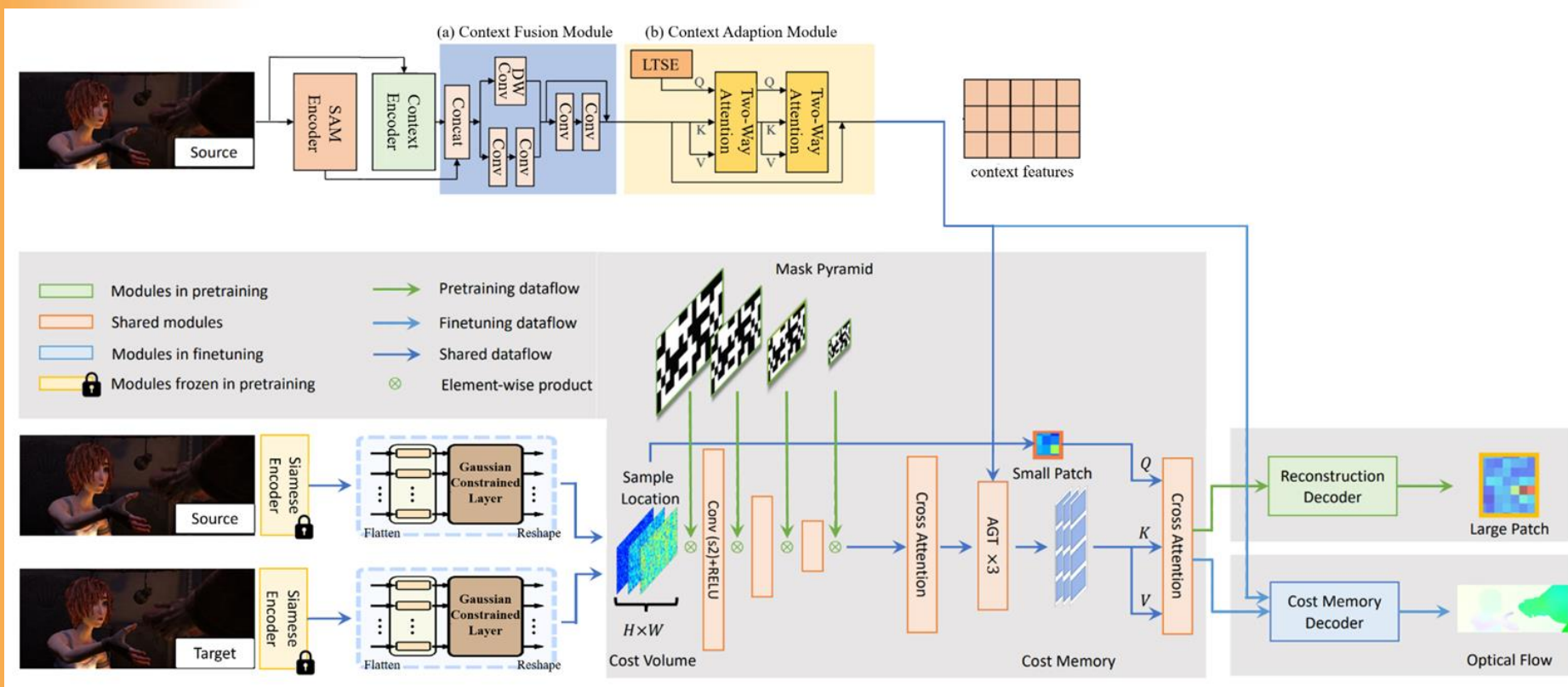


Fig. 2. Framework of the proposed approach.

Proposed Approach (cont.)

- The pretrained SAMFlow architecture containing context fusion module (CFM) and context adaption module (CAM) is shown in Fig. 3.

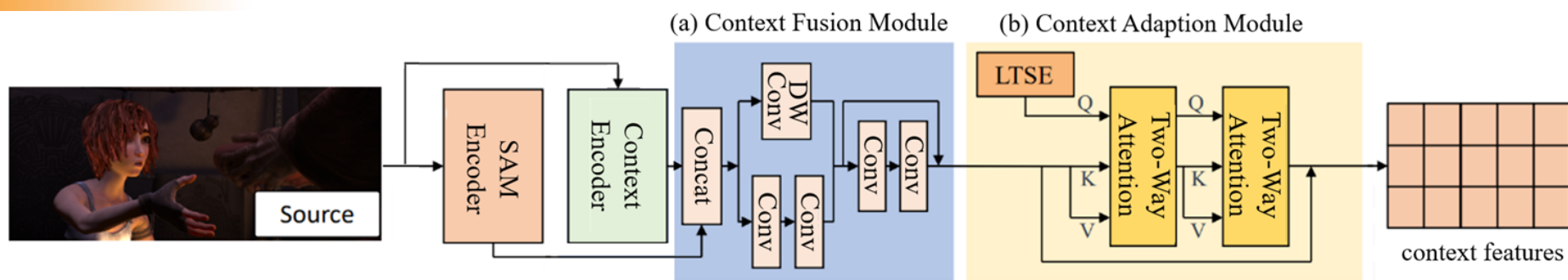


Fig. 3. The pretrained SAMflow architecture.

Proposed Approach (cont.)

- Context fusion module (CFM) fuses outputs of SAM encoder and context encoder to prevent task mismatch of local features and improves the effectiveness of SAMFlow.
- CFM will combine global understanding extracted by SAM encoder and local clues extracted by both SAM encoder and context encoder.

Proposed Approach (cont.)

- CFM combines SAM encoder and context encoder to extract more comprehensive features:

$$\Phi_S = E_S(I), \Phi_F = E_F(I)$$

$$\Phi_{\parallel} = \Phi_S \parallel \Phi_F$$

$$\bar{\Phi}_C = \text{Conv}_2(\text{Conv}_1(\Phi_{\parallel})) + \Delta(\Phi_{\parallel})$$

$$\Phi_C = \text{Conv}_4(\text{Conv}_3(\Phi_{\parallel})) + \bar{\Phi}_C$$

Proposed Approach (cont.)

- Context adaption module (CAM) can better utilize the task specific knowledge to accomplish task adaption of optical flow estimation.
- Learned task-specific embedding (LTSE) tokens are employed to store some task specific priors of optical flow, and two two-way attention (TWA) blocks are employed to inject those priors into context features for adaption.

Proposed Approach (cont.)

- LTSE containing a set of learnable offsets of shape 3×256 will be automatically optimized during the training process.
- As shown in Fig. 4, each TWA block contains four sub-blocks:

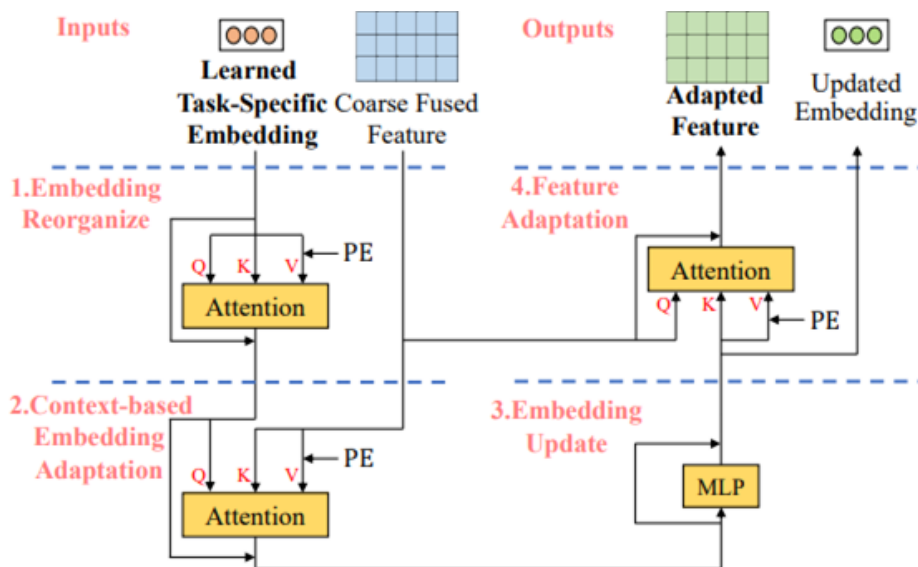


Fig. 4. Architecture of TWA block.

Proposed Approach (cont.)

- A self-attention layer (1st sub-block) is used to reorganize the embedding of optical flow estimation task Ω_T (the output of LTSE):

$$\bar{\Omega}_T = \Omega_T + \text{Att}_1(\Omega_T, \Omega_T, \Omega_T + PE)$$

where PE denotes positional embedding.

- A cross-attention layer (2nd sub-block) is used to adapt the embedding with context features:

$$\hat{\Omega}_T = \bar{\Omega}_T + \text{Att}_2(\bar{\Omega}_T, \Phi_C, \Phi_C + PE)$$

where Φ_C denotes the output of CFM.

Proposed Approach (cont.)

- A multi-layer perceptron (MLP) (3rd sub-block) is used to update the query:

$$\Omega_U = \hat{\Omega}_T + MLP(\hat{\Omega}_T)$$

- A cross-attention layer (4th sub-block) is used to adapt the context features for optical flow estimation:

$$\Phi_C^A = \Phi_C + Att_3(\Phi_C, \Omega_U, \Omega_U + PE)$$

Proposed Approach (cont.)

- Gaussian constrained layer (GCL) fortified with a learnable Gaussian kernel, naturally prioritizes the local surroundings of each pixel, thus effectively enhances the feature discrimination for cross-frame matching.
- GCL can obtain the fine-grained structural information and locally discriminative representations for feature matching.

Proposed Approach (cont.)

- As shown in Fig. 5, GCL is integrated into FlowFormer++ encoder, which processes extracted features before generating the cost volume.

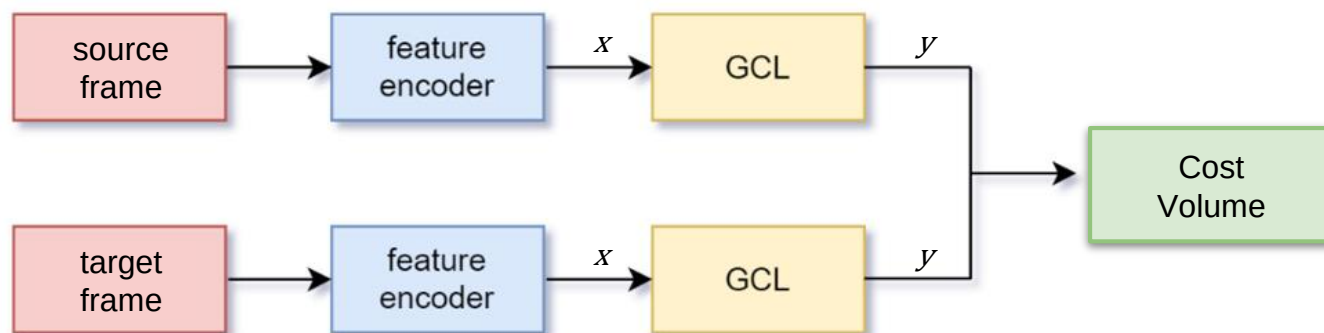


Fig. 5. Architecture of the proposed cost volume encoder.

Proposed Approach (cont.)

- For input feature x , the output y of GCL is defined as:

$$\hat{x} = GCA(LN(x)) + x$$

$$y = FFN(LN(\hat{x})) + \hat{x}$$

where GCA (Gaussian constrained attention) is a task-specific local operation:

$$Q_i = L_i^Q(x'), K_i = L_i^K(x'), V_i = L_i^V(x'),$$

$$h_i = \mathcal{G}_A(Q_i, \bar{K}_i, \bar{V}_i),$$

$$H = \text{Concat}(h_1, h_2, \dots, h_n),$$

$FFN(\cdot)$ denotes feed-forward network, and $LN(\cdot)$ denotes layer normalization.

Proposed Approach (cont.)

- In this study,

$$\mathcal{G}_A(Q, \bar{K}, \bar{V}) = \text{Softmax}(G_{lr} \cdot (Q\bar{K}^T)/\sqrt{d}) \cdot \bar{V},$$

$h_p, p = 1, 2, \dots, n$ is defined as

$$h_p = \text{Softmax}(G_{lr} \cdot (Q_{(p)}\bar{K})_{N(p)}^T/\sqrt{d}) \cdot \bar{V}_{N(p)},$$

and $\text{Concat}(\cdot, \cdot, \dots, \cdot)$ denotes concatenation operation.

Experimental Results

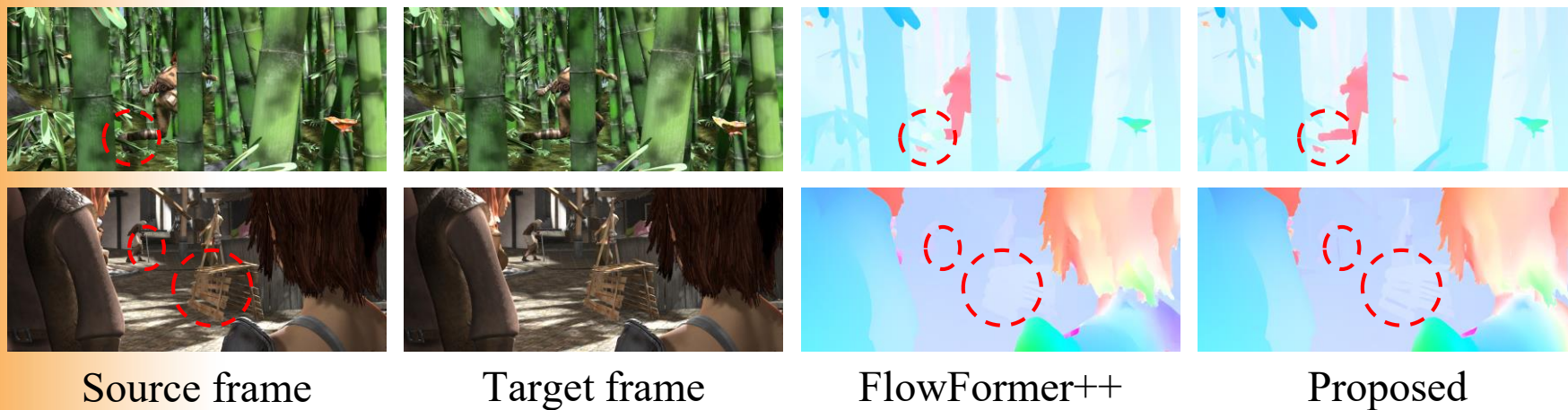


Fig. 6. Visual comparisons between FlowFormer++ and the proposed approach on two samples of Sintel dataset.

Experimental Results (cont.)



Fig. 7. Visual comparisons between FlowFormer++ and the proposed approach on one video sequence of Sintel dataset.

Experimental Results (cont.)

Table 1. In terms of average EPE and F1-all, performance comparisons between FlowFormer++ and the proposed approach on Sintel and KITTI datasets.

	Sintel		KITTI	
metrics approaches	EPE-clean	EPE-final	EPE-F1	F1-all
FlowFormer++	0.90	2.30	3.93	14.13
Proposed	0.85	2.05	3.96	13.86

Concluding Remarks

- In this study, a modified optical flow estimation approach with SAM and GCL is proposed.
- Based on the experimental results obtained in this study, in terms of two performance metrics and subjective evaluation, the performance of the proposed approach is better than that of FlowFormer++.

Thanks for your attention!